

Abu Bakar Yasir

Full-Stack & AI Engineer
NEXT.JS • FASTAPI • LLM
SYSTEMS

CONTACT

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Lahore, Pakistan

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TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL

AI / ML: RAG, LLM Integration (OpenAI, Anthropic Claude, Groq, Gemini), Multi-Agent Orchestration, Embeddings, Prompt Engineering, TensorFlow, PyTorch

Backend: FastAPI, Django, Django REST Framework, Node.js, Express.js, REST APIs, WebSockets

Frontend: Next.js, React.js, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Supabase, ChromaDB, Qdrant, Neo4j

DevOps & Tools: Docker, Railway, Vercel, Redis, Git, GitHub, Postman

EDUCATION

BS Computer Engineering
COMSATS University Islamabad (CUI)
Sep 2022 – Jun 2026 · Lahore, Pakistan
CGPA: 3.29 / 4.0

PROFESSIONAL SUMMARY

Full-Stack & AI Engineer with **1+ year of professional experience** shipping production-grade web applications and AI systems end-to-end. Builds **multi-tenant SaaS platforms**, Retrieval-Augmented Generation (RAG) pipelines, and multi-agent LLM orchestration with Next.js, FastAPI, and Django REST Framework. Strong in REST API design, secure authentication (JWT/RBAC), vector databases, and cloud deployment.

PROFESSIONAL EXPERIENCE

Full-Stack & AI Engineer — Spiral Lab

Remote / Lahore, Pakistan | April 2025 – Present

- Shipped production-grade, full-stack applications end-to-end across **3+ products** using Next.js, FastAPI, Django, Node.js, and Supabase/PostgreSQL, owning backend, AI layers, async pipelines, and frontend.
- Built AI-driven systems including **RAG pipelines** and **multi-agent LLM orchestration** (OpenAI, Anthropic Claude, Groq), integrating vector databases (Qdrant, ChromaDB) for low-latency semantic retrieval.
- Engineered secure, **multi-tenant architectures** with JWT authentication, RBAC, row-level security, rate limiting, and audit logging, plus async pipelines (Django-Q2, Redis, WebSockets) deployed to Railway and Vercel with CI auto-deploy.

KEY PROJECTS

Magnai — Vertical AI SaaS for UK Public Affairs **LIVE**

Python · Django REST Framework · PostgreSQL · Anthropic Claude Agent SDK · OpenAI · Django-Q2 · Redis · Railway

- Built and deployed a **multi-tenant Django SaaS** with a multi-agent AI system (orchestrator plus parliament / media / Twitter sub-agents, each with its own tool registry) monitoring UK, Scottish, and Welsh data to auto-generate AI-authored intelligence reports.
- Developed robust async monitoring pipelines (Django-Q2 and Redis) with race-condition protection, a security layer (rate limiting, anomaly detection, CSP), and a stakeholder-mapping engine with relevance scoring.

Lynxflow-Health — Healthcare Management System (SaaS) **LIVE**

Next.js · Node.js · Supabase (PostgreSQL) · REST APIs

- Built a responsive, role-based healthcare platform (Clinical and Super Admin) with **row-level security (RLS)** and HIPAA-aligned data handling; integrated Tenovi smart-pillbox hardware via webhooks for medication-adherence tracking, plus in-platform device ordering and a vulnerability-scanning bot.

neuroMedica — Medical RAG System for Clinical Decision Support **LIVE**

Next.js · TypeScript · FastAPI · Qdrant · Neo4j · PubMedBERT · Anthropic Claude · TensorFlow · PyTorch · Docker

- Designed a **hybrid-retrieval RAG pipeline** fusing dense (PubMedBERT / Qdrant), sparse (BM25), and knowledge-graph (Neo4j / UMLS) retrieval via Reciprocal Rank Fusion, with a CRAG self-evaluation loop to cut hallucination for clinical safety.
- Built deep-learning diagnostic modules: a Keras CNN for ECG arrhythmia classification (R-peak detection via neurokit2) and a DenseNet-121 (PyTorch) multi-label chest X-ray pathology classifier; deployed full stack via Docker on Railway.

Slack-Bounty-Bot — Real-Time Replit Bounty Notifier **GITHUB**

Python · Flask · Firecrawl API · Slack Webhooks · Vercel

- Built a Python microservice that scrapes and filters high-value Replit bounties in real time and pushes automated Slack alerts; deployed as a Flask API on Vercel with CRON-based continuous monitoring.